Windcave Hosted –PX Pay 2.0 Integration Guide

Version 3.6

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Contents

1 Overview

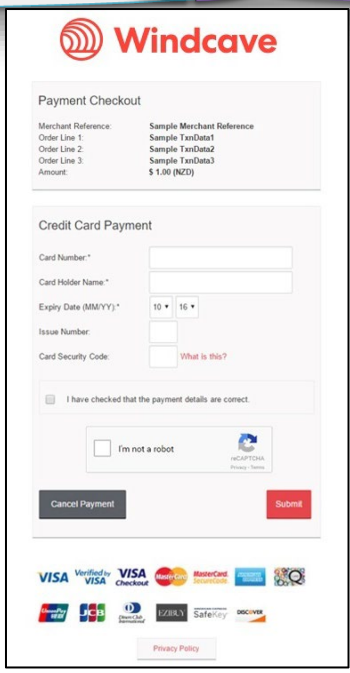
The PX Pay 2.0 interface is a platform independent Hosted Payments Page (HPP) solution provided by Windcave. The HPP provides a solution for the capturing credit card information securely without exposing the merchant to the sensitive data.

This is achieved by allowing the card holder to enter their card details into a page which is hosted by Windcave rather than the merchants own website. The major advantage of this approach is that the merchant does not see, and is not aware of, the card number at any point in the process. This is beneficial from a PCI DSS standpoint because the scope of PCI DSS requirements is likely to be reduced.

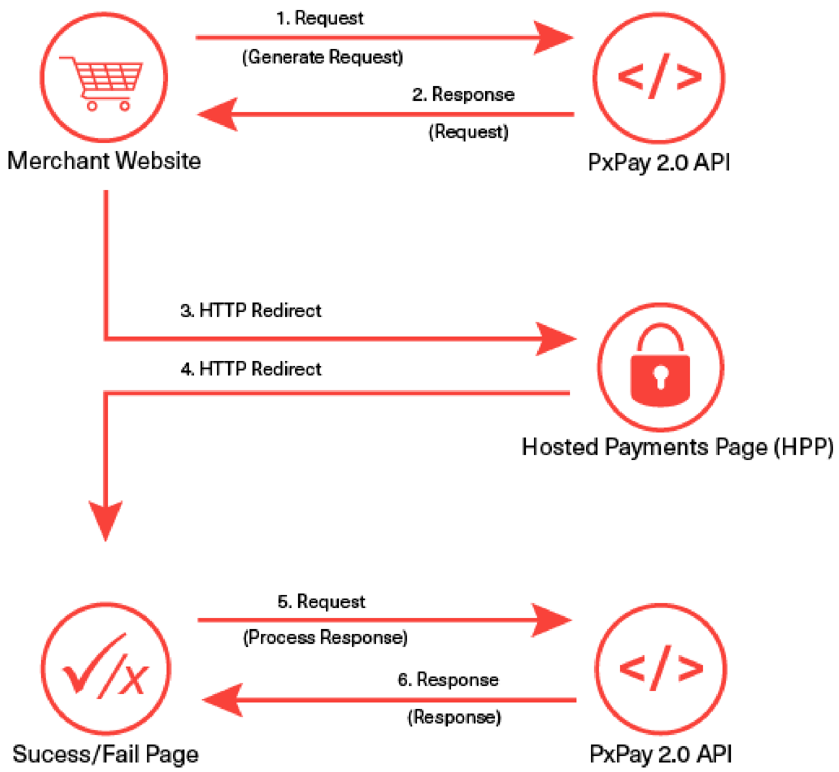
PCI DSS (Payment Card Industry Data Security Standard) is a set of comprehensive requirements created by card issuers American Express, Discover Financial Services, JCB International, MasterCard and Visa to ensure the security of credit card data online. All merchants, whether small or large, need to be PCI compliant. Windcave is registered as a PCI DSS compliant service provider; therefore a payment page solution hosted by Windcave meets all PCI DSS requirements.

1.1 Key features

A demonstration of PX Pay 2.0 can be found online at https://demo.windcave.com/SandboxMain.aspx



2 How It Works



1. To process a transaction, PX Pay 2.0 allows merchants to send XML requests to Windcave® via HTTPS posts to https://sec.windcave.com/pxaccess/pxpay.aspx. PxPay Username & PxPay Key is required too.

2. Windcave® responds with a unique URI (encrypted URL) for an SSL secure payments page.

3. The merchant shopping cart uses the returned URI to redirect the customer to the secure Windcave hosted payments page.

4. The customer will be prompted to enter their credit card details and complete the transaction. The transaction is then sent to the merchant bank for authorisation. The result is displayed and the user is automatically redirected back to the merchant's website (success or fail page); i.e.https://demo.windcave.com/SandboxSuccess.aspx?result=0000840000185376f1519ff80a5ccd54&userid= SampleUserId

5. You take the "result" parameter value in the URL string i.e. 0000840000185376f1519ff80a5ccd54 along with the PX Pay username and PX Pay key; to send the response request (ProcessResponse) to Windcave and receive the XML response back.

6. The transaction results and other transaction details are decrypted and sent back to the merchant as a standard XML response. NOTE: In case a blank XML response is returned, please retry the ProcessResponse twice with an interval of 2 seconds.

3 Basic Communication

Character data sent via PX Pay 2.0 must be well formed XML.

The XML document must contain the required opening and closing tags that contains the whole document i.e. the root element.

Example: When generating the input XML document to begin a transaction request, the following GenerateRequest opening and closing tags must be present.

<GenerateRequest>
...
</GenerateRequest>

All tags must be nested properly. There must be an opening and a closing tag for all elements and the tags cannot overlap.

Example: Closing tags not complete.

</AmountInput - *has no closing angle bracket, therefore the tag is not complete.*

</AmountInput) - *has a wrong closing bracket, therefore the tag is not complete.*

The XML tags are case sensitive and unique. If a tag is submitted which is not recognized by Windcave and is not a required element, it will be ignored and will not be returned in the response. If the tag is for a required element, an error may occur and a response code will be returned.

Example: If the AmountInput tag was sent with a lowercase "i" instead of an uppercase "I" and error will occur the response code "IU – Invalid Amount" will be returned

<Amountinput>1.00</Amountinput> - *Incorrect*

<AmountInput>1.00</AmountInput> - *Correct*

If there is a possibility that a value will contain **invalid characters** (such as '&' and '>' in the cardholder name), please format the value using "HtmlEncoding", otherwise Windcave will be unable to read the XML and will return an error (i.e. "Not acceptable input XML").

Example, the following is invalid XML:

```
<GenerateRequest>

<TxnData1>Bill & Son</TxnData1>

<MerchantReference>Abc >> 123</MerchantReference>

</GenerateRequest>
```

The following is how it should be formatted.

```
<GenerateRequest>

<TxnData1>Bill &amp; Son</TxnData1>

<MerchantReference>Abc &gt;&gt; 123</MerchantReference>

</GenerateRequest>
```

4 Integration Methods

Generally merchants implement a Windcave hosted payment page solution in one of two ways; either redirecting the user and their entire browser to the payment page or by presenting the payment page within an inline frame embedded in a page on their website.

4.1 Mobile Device

The PxPay 2.0 interface can also be integrated on a native mobile application as a payment method. The mobile application may utilise a webpage component to view the hosted payment page over HTTPs. The mobile platform being integrated with PxPay 2.0 should support HTTPS posts and XML data exchange.

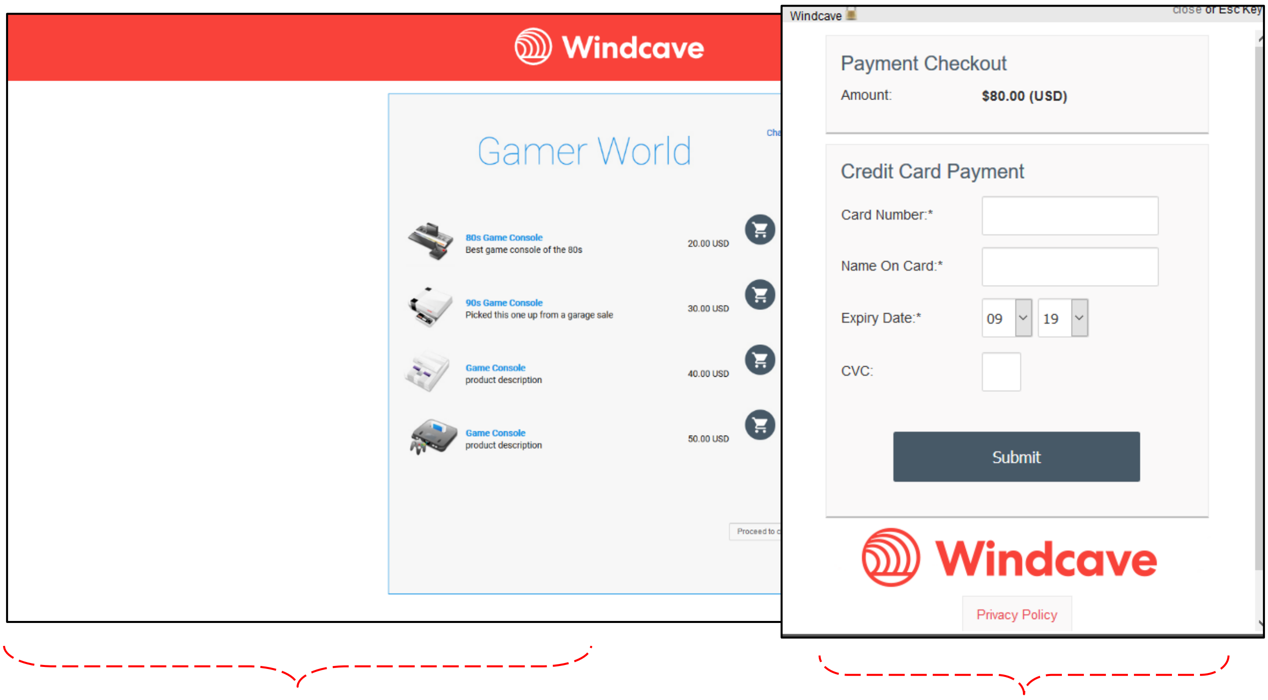
The PxPay 2.0 payment automatically switches to a mobile format for mobile devices by default. If the PxPay 2.0 payment page is not responsive to the mobile device's screen as expected, please ensure the user agent string that the device's web browser is reporting is mobile specific within the first 255 characters of the string.

If the hosted payment page is still not responsive on the mobile device screen, please note the following:

- Please email devsupport@windcave.com and quote the exact user agent string of the relevant mobile device(s) used to send the transaction request.
- Also if a mobile web browser is used to redirect to the hosted payment page, please specify the exact mobile web browser and the version.

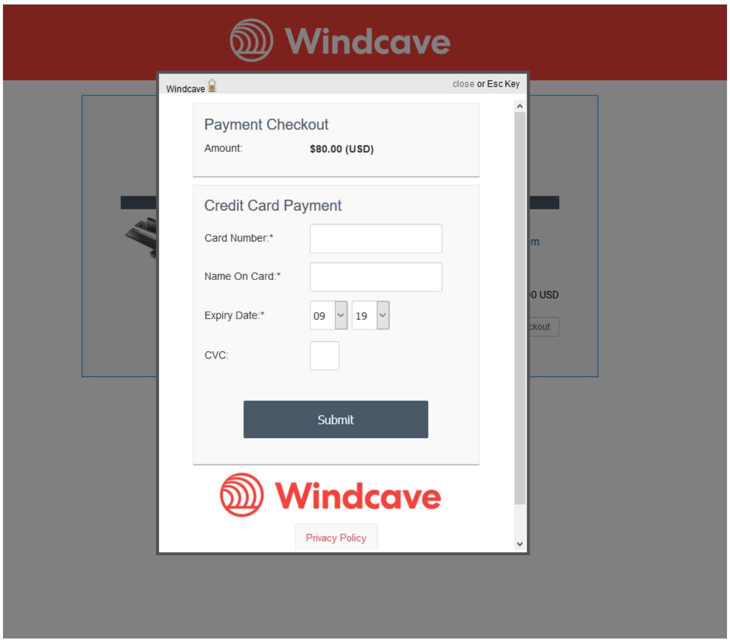
4.2 Redirect

The redirect integration method involves directing the user away from the merchant website to a Windcave-hosted page for the purposes of collecting credit card details. Once credit card details have been collected and a transaction processed the user is directed back to the merchant website. The image below demonstrates a payment page accessed using the redirection method.

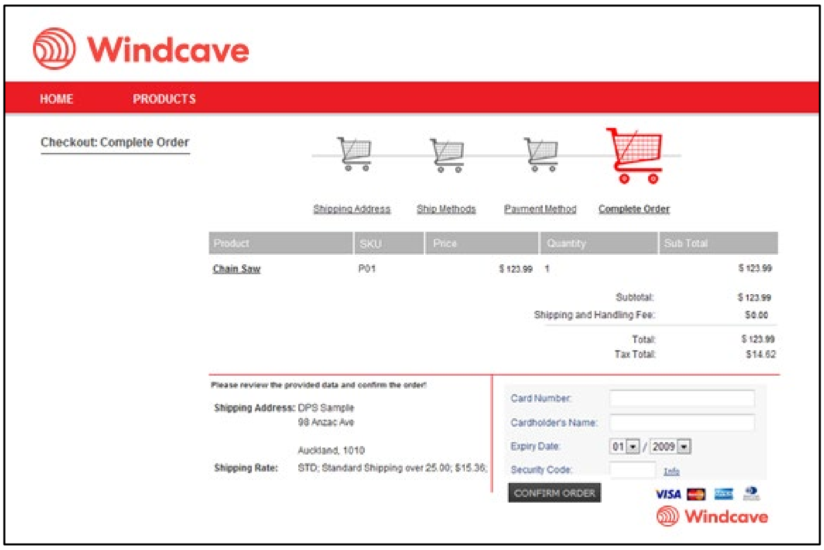


4.3 Iframe – Embedded Hosted Payment Page

The iframe integration method involves presenting the Windcave hosted payment page within the merchant website inside a frame. The iframe content can either be presented as the page loads or asynchronously (outside the normal page request flow) based upon user interaction. Note that this method of integration may increase the scope of applicable PCI-DSS requirements. Please speak to your acquirer to confirm their position on this particular implementation of the Windcave hosted payment page. Please note that the iframe window must be large enough to present the entire 400 pixel width by 470 pixel length for 3DS ACS page to display without any scrolling required.The images below demonstrate the iframe method of integration.



Windcave Hosted Payments Page



Windcave Hosted Payment Page (HPP)
embedded in an iframe

5 Preparation

To begin integration testing you will need the following:

- Windcave® PX Pay 2.0 development accounts – Contact our Ecommerce sales team to request a Dev account (Please refer to: https://www.windcave.com/contact), or apply online at https://sec.windcave.com/pxmi/apply.
- PX Pay 2.0 interface technical specification - www.windcave.com/developer-e-commerce-hosted-pxpay
- PX Pay 2.0 sample code - http://www.windcave.com/Downloads/PxPay2_SampleCode.zip

5.1 PX Pay 2.0 development account

A PX Pay 2.0 development account is usually setup within 1-3 business days. Each test account will be assigned to the Windcave test environment which simulates a connection to the merchant bank. To access the PX Pay 2.0 account, a UserId and Key will be provided.

Example: PxPayUserId: Sample2_Dev
PxPayKey: abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890

All PX Pay 2.0 accounts also come with a Payline account. Developers can use Payline to track down their test transactions, process transactions manually, and generate transaction reports. To access the Payline account, use the PXPayUserId along with a unique alphanumeric password setup just for Payline.

Payline login URL: https://sec.windcave.com/pxmi3/logon.

Example: Payline Username: Sample2_Dev (Same as PxPay2UserId)
Payline Password: abcd1234

5.2 PX Pay 2.0 sample code

Sample code can be provided in the following languages:

- PHP cURL
- PHP OpenSSL
- ASP.Net 3.5 (C#)
- ASP.Net 3.5 (VB)
- Java
- ColdFusion

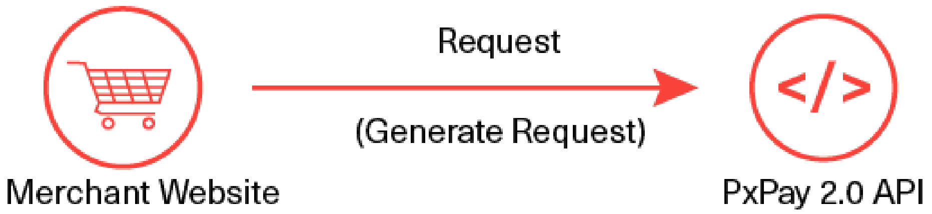
You will need to ensure the client library sending HTTP request is using TLSv1.2 for connectivity at all times.

All the sample codes can be downloaded from the link:

https://www.windcave.com/Downloads/PxPay2_SampleCode.zip

6 Transaction Request

GenerateRequest XML Document – To initiate a transaction the merchant posts the GenerateRequest to https://sec.windcave.com/pxaccess/pxpay.aspx




```xml
<GenerateRequest>
<PxPayUserId>SampleUserId</PxPayUserId>
<PxPayKey>abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890</PxPayKey>
<TxnType>Purchase</TxnType>
<AmountInput>1.00</AmountInput>
<CurrencyInput>NZD</CurrencyInput>
<MerchantReference>Purchase Example</MerchantReference>
<TxnData1>John Doe</TxnData1>
<TxnData2>0211111111</TxnData2>
<TxnData3>98 Anzac Ave, Auckland 1010</TxnData3>
<EmailAddress>SampleUserId@windcave.com</EmailAddress>
<TxnId>ABC123</TxnId>
<BillingId>BillingId123xyz</BillingId>
<EnableAddBillCard>1</EnableAddBillCard>
<RecurringMode>single</RecurringMode>
<UrlSuccess>https://demo.windcave.com/SandboxSuccess.aspx</UrlSuccess>
<UrlFail>https://demo.windcave.com/SandboxSuccess.aspx</UrlFail>
<UrlCallback>https://InsertValidUrlForCallback</UrlCallback>
</GenerateRequest>
```

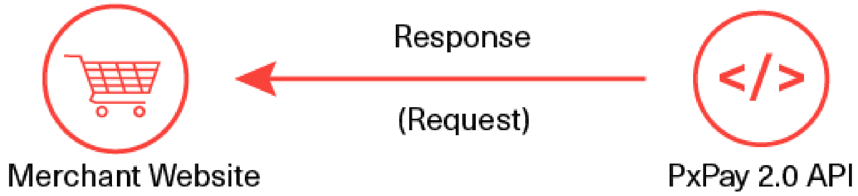
Note: Elements in blue text are optional

The following is a list of the inputs elements applicable for a GenerateRequest.

GenerateRequest (Input XML Document)

Input Element	**Required**	**Datatype**
PxPayUserId	Yes	Max 32 bytes
PxPayKey	Yes	Max 64 bytes
AmountInput	Yes	Max 13 characters
BillingId	No	Max 32 characters
CurrencyInput	Yes	Max 4 characters
EmailAddress	No	Max 255 bytes
EnableAddBillCard	No	Max 1 character
RecurringMode	No	String
MerchantReference	No	Max 64 bytes
DpsBillingId	No	Max 16 characters
TxnData1	No	Max 255 bytes
TxnData2	No	Max 255 bytes
TxnData3	No	Max 255 bytes
TxnType	Yes	Max 8 Characters
TxnId	Yes	Max 16 bytes
UrlFail	Yes	Max 255 bytes
UrlSuccess	Yes	Max 255 bytes
UrlCallback	No	Max 255 bytes
Opt	No	Max 64 bytes
ClientType	No	String
ForcePaymentMethod	No	String
DebtRepaymentIndicator	No	Number
InstallmentNumber	No	Number
InstallmentCount	No	Number

Request XML Document - Once the GenerateRequest has been processed a Request will be returned.



The URI returned can then be used to redirect the customer to the Windcave Hosted Payments Page.

The following is a list of the output elements applicable for a Request.

```
<Request valid="1">

<URI>https://sec.windcave.com/pxmi3/EF4054F622D6C4C1B4F9AEA59DC91CAD3654CD60ED7ED04110
CBC402959AC7CF035878AEB85D87223</URI>

</Request>
```

Request (Output XML Document)

Output Element	**Datatype**
valid [Attribute]	1 character
URI	String

7 Transaction Response

ProcessResponse XML Document – Once the user has submitted their credit card information and the transaction has been processed, the merchant's UrlSuccess or UrlFail or UrlCallback (if set in the request) will receive an FPRN with the GET HTTP message with the result value that needs to extracted and requested as below to obtain the transaction outcome and details.
To extract the transaction details from the encrypted URI string, the merchant sends the encrypted result string as Response value with their PXPayUsername and PXPaykey in the ProcessResponse (Input XML Document).

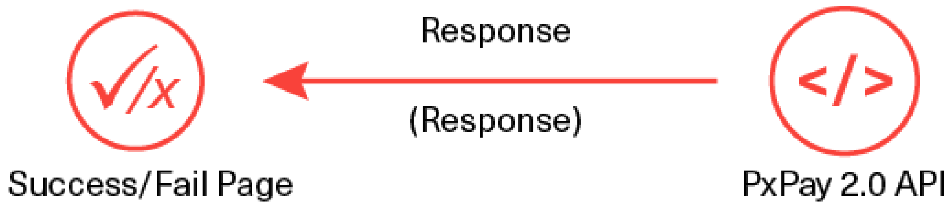
The following is a list of the inputs elements applicable to the ProcessResponse. A full description of all available elements can be found on *Element Descriptions* section.
. Example:

```
<ProcessResponse>

<PxPayUserId>SampleUserId</PxPayUserId>

<PxPayKey>abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890</PxPayKey>

<Response>00008400000185376f1519ff80a5ccd54</Response>

</ProcessResponse>
```

ProcessResponse (Input XML Document)

Input Element	Required	Datatype
PxPayUserId	Yes	Max 32 bytes
PxPayKey	Yes	Max 64 bytes
Response	Yes	String

Response XML Document - The Response will return the decrypted transaction results for the merchant to interpret.



The following is a list of the output elements applicable to the Response. A full description of all available elements can be found on *Element Descriptions section*.

Response (Output XML Document)

Output Element	Datatype			Output Element	Datatype
valid [Attribute]	1 character			TxnData1	Max 255 bytes
AmountSettlement	Max 13 characters			TxnData2	Max 255 bytes
AuthCode	Max 22 characters			TxnData3	Max 255 bytes
CardName	Max 16 bytes			TxnType	Max 8 Characters
CardNumber	Max 16 bytes			CurrencyInput	Max 4 characters
DateExpiry	Max 4 characters			MerchantReference	Max 64 bytes
DpsTxnRef	Max 16 bytes			ClientInfo	String
Success	Long			TxnId	Max 16 bytes
ResponseText	Max 32 bytes			EmailAddress	Max 255 bytes
DpsBillingId	Max 16 characters			BillingId	Max 32 characters
CardHolderName	Max 64 bytes			TxnMac	Max 32 characters
CurrencySettlement	Max 4 characters			CardNumber2	Max 16 characters
PaymentMethod	String			Cvc2ResultCode	1 character

8 Element Descriptions

PxPayUserId (input) Datatype: BSTR Max 32 bytes
The PxPayUserId is a unique username to identify your customer account. This username will be setup by our Windcave® Activations team and issued with your development account. The standard format used for Dev PxPayUserId is 'merchantname_dev' e.g. storename_dev.

PxPayKey (input) Datatype: BSTR Max 64 bytes
The PxPayKey is a unique 64 character key to identify customer account and used to encrypt the transaction request with 3DES to protect the transaction information. This key will be setup by our Windcave® Activations team and issued with your development account.

AmountInput (input) Datatype: BSTR Max 13 characters
The AmountInput field specifies the total Purchase or Auth amount.

Format is d.cc where d is dollar amount (no currency indicator) and cc is cents amount. For example, $1.80 (one dollar and eighty cents) is represented as "1.80", not "1.8". A string value is used rather than the conventional currency datatype to allow for easy integration with web applications. The maximum value allowable is "999999.99" however acquirer or card limits may be lower than this amount.

When submitting transactions for currencies with no decimal division of units such as the Japanese Yen (JPY), the AmountInput must be in an appropriate format e.g. "1000" for 1000 Yen. If "1000.00" is submitted in AmountInput with CurrencyInput set to JPY, an error will occur and response code "IU" will be returned.

BillingId (input/output) Datatype: BSTR Max 32 characters
If a token based billing transaction is to be created, a BillingId may be supplied. This is an identifier supplied by the merchant application that is used to identify a customer or billing entry and can be used as input instead of card number and date expiry for subsequent billing transactions. Token Billing is discussed in detail on Token Billing Section.

CurrencyInput (input/output) Datatype: BSTR Max 4 characters
The CurrencyInput field is used to specify the currency to be used e.g. "NZD" or "AUD". Possible range of currencies:

AUD	Australian Dollar		HKD	Hong Kong Dollar		SGD	Singapore Dollar
BND	Brunei Dollar		INR	Indian Rupee		THB	Thai Baht
CAD	Canadian Dollar		JPY	Japanese Yen		TOP	Tongan Pa'anga
CHF	Switzerland Franc		KWD	Kuwait Dinar		USD	United States Dollar
EUR	Euro		MYR	Malaysian Ringgit		VUV	Vanuatu Vatu
FJD	Fiji Dollar		NZD	New Zealand Dollar		WST	Samoan Tala

FRF	French Franc		PGK	Papua New Guinean Kina		ZAR	South African Rand
GBP	United Kingdom Pound		SBD	Solomon Islander Dollar			

EmailAddress (input/output) Datatype: BSTR Max 255 bytes
The EmailAddress field can be used to store a customer's email address and will be returned in the transaction response. The response data along with the email address can then be used by the merchant to generate a notification/receipt email for the customer. This is an optional field.

EnableAddBillCard (input) Datatype: BSTR Max 1 character
The EnableAddbillCard field is used if subsequent billing is required. Setting this field to "1" will cause Windcave to store the credit card details for future billing purposes. A DpsBillingId (Windcave generated) or BillingId (merchant generated) is used to reference the card. More information on Token/Recurring billing can be found on Token Billing Section.

MerchantReference (input/output) Datatype: BSTR Max 64 bytes
The Merchant Reference field is a free text field used to store a reference against a transaction. The merchant reference allows users to easily find and identify the transaction in Payline transaction query and Windcave reports. The merchant reference is returned in the transaction response, which can be used interpreted by the merchant website. Common uses for the merchant reference field are invoice and order numbers. This is an optional field.

ForcePaymentMethod (input) Datatype: String
Allows to force the specific payment method on the hosted payment page to use if the PxPayUserId has multiple payment methods configured. Only one of the below string values should be set. Please note certain payment methods such as Account2Account, Alipay, UPOP5 (China UnionPay) and WeChat, only accept the Purchase transaction type.
Any other values are deemed invalid according to the response.
If this field is not set then by default, multiple payment method options configured would be available on the hosted payment page.

- Account2Account
- Alipay
- AmexExpressWallet
- Card
- GECard
- MasterPass
- MonerisIOP
- Oxipay
- PayPal
- UPOP5
- UPOP
- VisaCheckout
- WeChat

PaymentMethod (output) Datatype: String
The final transaction response can contain the PaymentMethod field to indicate the specific payment method used for the transaction. If this field does not exist in the final transaction response then a card payment method was used by default.

ClientType (input) Datatype: BSTR
Allows the Client type to be passed through at transaction time. Please note the feature may need to be enabled for the PxPay 2.0 user. Recommended values for PxPay 2.0:
Internet = I
Recurring = R (if supported by your MID at your merchant bank expiry date will not be validated)
Please note the single letter input is case sensitive.
Please submit the full word or a single letter.

RecurringMode (input) Datatype: BSTR Max 255 bytes
In the RecurringMode request field, please set one of the card storage reason as the string listed below:
When **tokenising the card**:

- recurringinitial
- installmentinitial
- credentialonfileinitial
- unscheduledcredentialonfileinitial

When **rebilling the card** with a token returned on tokenising, set RecurringMode as one of:

- credentialonfile
- unscheduledcredentialonfile
- incremental
- installment
- recurring
- recurringnoexpiry
- resubmission
- reauthorisation
- delayedcharges
- noshow

Note: The RecurringMode string value should be set based on the merchant's business case or use case. Further details in Token Billing section.

InstallmentNumber (input) Datatype: Number
Number based value that is used to indicate the current payment number for an installment transaction. For example, if the consumer is making payment 1 of 12, then this value should be set to 1. Only used for installment based payments.

InstallmentCount (input) Datatype: Number
The number value is used to indicate the total number of payments for an installment transaction. For example, if the consumer is making 12 installment payments for total payment, then this value should be set to 12. Only used for installment based payments.

DebtRepaymentIndicator (input) Datatype: Number
Only send this field and set it to 1 to indicate that a debt repayment transaction is to be processed. Otherwise it will be set to 0 by default.

TxnData1, TxnData2, TxnData3 (input/output) Datatype: BSTR Max 255 bytes (each field)
The TxnData fields are free text fields that can be used to store information against a transaction. This can be used to store information such as customer name, address, phone number etc. This data is then returned in the transaction response and can also be retrieved from Windcave reports. This is an optional field.

Note: storing large strings of information in the TxnData fields can cause the URI returned in the Request (Response) to exceed the max URL length for some browsers.

TxnType (input/output) Datatype: BSTR Max 8 characters
The TxnType can either be "Purchase" or "Auth".

A Purchase transaction type processes a financial transaction immediately and funds are transferred at next settlement cut-off. Basically a Purchase transaction debits the card in real time and if you need to cancel the transaction you will need to do a refund. An example of the Purchase transaction can be found on Purchase Transaction Example section.

The Auth transaction on the other hand, transfers no money. An Auth transaction type verifies that funds are available for the requested card and amount and reserves the specified amount. The amount is reserved for a period of up to 7 days (depending on merchant bank) before it clears. If you do not wish to take money from the customer you do nothing, and the reservation on the money will expire after 7 days. If you wish to process the payment a Complete transaction is sent to finalize the funds transfer.

For more information on the Auth and Complete transaction can be found on Auth Transaction Example section.

TxnId (input/output) Datatype: BSTR Max 16 bytes
On GenerateRequest XML it is a mandatory request field. Should always contain a unique, merchant application generated value that uniquely identifies the transaction. Used by Windcave® to check for a duplicate transaction generated from Merchant web site. If a duplicate TxnId in the request is detected per PxPay account, the hosted payment page URI will not be generated and an error will be returned in the XML response of the GenerateRequest XML request and the merchant site should be informed of the result. Please ensure that the value is always unique per transaction request.

UrlFail (input) Datatype: BSTR Max 255 bytes
Url of the page to redirect to if transaction failed. Please ensure the URL starts with the protocol, for example: "https:..." or "http:...". Query string characters are permitted. However percent-encoding the URL is not permitted. Please avoid using any custom ports for this URL.

UrlSuccess (input) Datatype: BSTR Max 255 bytes
Url of the page to redirect to if transaction successful. Please ensure the URL starts with the protocol, for example: "https:..." or "http:...". Query string characters are permitted. However percent-encoding the URL is not permitted. Please avoid using any custom ports for this URL.

UrlCallback (input) Datatype: BSTR Max 255 bytes
Externally accessible URL of page or resource to receive the notification on the conclusion of the HPP session when the transaction outcome is determined. Please ensure the URL starts with the protocol, for example: "https:..." or "http:...". Query string characters are permitted. Please avoid using any custom ports for this URL.

Opt (input) Datatype: BSTR Max 64 bytes
This optional parameter can be used to set a timeout value for the hosted payments page or block/allow specified card BIN ranges.

Timeout – A timeout (TO) can be set for the hosted payments page, after which the payment page will timeout and no longer allow a payment to be taken. Note: The default timeout of the created hosted payment page session is 72 hours. A specific timeout timestamp can also be specified via the request in Coordinated Universal Time (UTC). The value must be in the format "TO=yymmddHHmm" e.g. "TO=1010142221" for 2010 October 14th 10:21pm.
The merchant should submit the timeout value of when the payment page should timeout. One approach is to find the time (UTC) from when the GenerateRequest input XML document is generated and add on how much time you wish to give the customer before the payment page times out.

Example:

1. Customer finishes last stage of checkout process and clicks proceed to be redirected to the hosted payments page.
2. GenerateRequest Input XML document is generated with the required fields and the Opt field is populated with a timeout "TO=yymmddHHmm" that is 10 minutes later than the current time.
3. When the customer is redirected to the hosted payments page they will have 10 minutes to complete the payment before the page times out.

If the page times out the customer will be prompted with a "Declined – Transaction timed out" response, indicating that the page has timed out and payment was not processed. The customer will then be returned to the merchant's UrlFail page.

This feature is useful for merchants who wish control how much time a customer has to pay for the product e.g. concert tickets, where the merchant does not want people to be able to hold seats for an extended period of time due to limited availability.

Valid (output) Datatype: BSTR Max 1 character
The valid field indicates whether the initial request was valid or invalid i.e. "1" for valid and "0" for invalid.

URI (output) Datatype: BSTR
The URI field returns the URL including the encrypted transaction request that you will need to redirect the user to.

Response (input) Datatype: BSTR
The Response field should contain the encrypted URL response from Windcave, which can be obtained from the "result" parameter in the URL string that is returned to your response page.

AmountSettlement (output) Datatype: BSTR Max 13 characters
The amount of funds settled with your bank for the transaction.

AuthCode (output) Datatype: BSTR Max 22 characters
Authorisation code returned for approved transactions from the acquirer.

CardName (output) Datatype: BSTR Max 16 bytes
The card type used for the transaction. Please refrain from processing any logic based on the value of CardName as it is subject to change from Windcave. Please see examples of possible major card names below.

CardName	CardNumber	CardType
Visa	4111111111111111	<CardName>Visa</CardName>
MasterCard	5431111111111111	<CardName>MasterCard</CardName>
Amex	371111111111114	<CardName>Amex</CardName>
Diners	36000000000008	<CardName>Diners</CardName>

CardNumber (output) Datatype: BSTR Max 16 bytes
The card number used for the transaction. The full credit card number isn't shown, however the bin range is given (first 6 characters) depending on masking pattern set.

DateExpiry (output) Datatype: BSTR Max 4 bytes
The expiry date of the card used in the transaction.

DpsTxnRef (input/output) Datatype: BSTR Max 16 bytes
A DpsTxnRef is returned for every transaction. If the transaction was approved, DpsTxnRef can be used as input to a Refund transaction i.e. used to specify a transaction for refund without supplying the original card number and expiry date.

Success (output) Datatype: Long
Indicates success or failure of the transaction. A value of 0 indicates the transaction was declined or there was an error. A value of 1 indicates the transaction was approved.

ResponseText (output) Datatype: BSTR Max 32 bytes
Response Text associated with the response code of the transaction.

DpsBillingId (input/output) Datatype: BSTR Max 16 characters
When output, contains the Windcave generated BillingId. Only returned for transactions that are requested by the application with the EnableAddBillCard value is set to true indicating a token billing entry should be created. Information on Token Billing can be found on Token Billing section.
CardHolderName (output) Datatype: BSTR Max 64 bytes
The cardholder name as it appears on customer card.

CurrencySettlement (output) Datatype: BSTR Max 4 characters
Used to specify the currency that was used for the transaction: AUD, USD, NZD etc.

ClientInfo (output) Datatype: String
The IP address of the user who processed the transaction.

BillingId (input/output) Datatype: BSTR Max 32 characters
If a token based billing transaction is to be created, a BillingId may be supplied. This is an identifier supplied by the merchant application that is used to identify a customer or billing entry and can be used as input instead of card number and date expiry for subsequent billing transactions.

TxnMac (output) Datatype: BSTR Max 32 characters
Indication of the uniqueness of a card number.

CardNumber2 (output) Datatype: BSTR Max 16 characters
A token generated by Windcave when adding a card for recurring billing. CardNumber2 is a 16 digit number which conforms to a Luhn 'mod 10' algorithm and has a 1-to-1 relationship with the actual card number used. Please contact Windcave support if you would like to use this value.

Cvc2ResultCode (output) Datatype: BSTR 1 character
The result of CVC validation.

Response Code	Definition	Interpreting Response Codes
M	CVC matched.	You will want to proceed with transactions for which you have received an authorisation approval. A CVC match indicates the values provided matches the Issuing Banks details
N	CVC did not match.	You may want to follow up with the cardholder to verify the CVC value before completing the transaction, even if you have received an authorisation approval. The CVC details provided by the Cardholder do not match their Issuing Banks details
P	CVC request not processed.	Issuing Bank is unable to process CVC at this time
S	CVC should be on the card, but merchant has sent code indicating there was no CVC.	You may want to follow up with the cardholder to verify that the customer checked the correct location for the CVC. If the transaction is Approved you may also wish to consider not fulfilling the transaction
U	Issuer does not support CVC.	The card Issuing bank does not support CVC process.

9 Common Scenarios

This section provides examples on the common PX Pay 2.0 scenarios.

All examples can be replicated on https://demo.windcave.com/SandboxMain.aspx using our PX Pay 2.0 Sandbox tool (https://demo.windcave.com/Sandbox.aspx).

9.1 Purchase Transaction Example

A Purchase transaction type processes a financial transaction immediately and funds are transferred at next settlement cut-off. Basically a Purchase transaction debits the card in real time and if you need to cancel the transaction you will need to do a refund.

Note additionally, if the PxPayUserId has multiple payment method configured by our activations team then you can force a specific payment method with the ForcePaymentMethod field as specified in the section Element Descriptions. Please note certain payment methods only accept the Purchase transaction type.

Step 1 - GenerateRequest (Input XML Document): Post to Windcave PX Pay 2.0 API (https://sec.windcave.com/pxaccess/pxpay.aspx).

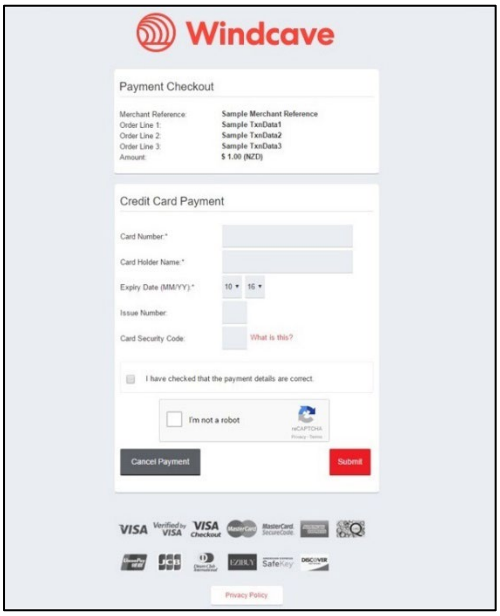
```
<GenerateRequest>
<PxPayUserId>SampleUserId</PxPayUserId>
<PxPayKey>abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890</PxPayKey>
<MerchantReference>Purchase Example</MerchantReference>
<TxnType>Purchase</TxnType>
<AmountInput>1.00</AmountInput>
<CurrencyInput>NZD</CurrencyInput>
<TxnData1>John Doe</TxnData1>
<TxnData2>0211111111</TxnData2>
<TxnData3>98 Anzac Ave, Auckland 1010</TxnData3>
<EmailAddress>SampleUserId@windcave.com</EmailAddress>
<TxnId>P03E57DA8A9DD700</TxnId>
<UrlSuccess>https://demo.windcave.com/SandboxSuccess.aspx</UrlSuccess>
<UrlFail>https://demo.windcave.com/SandboxSuccess.aspx</UrlFail>
</GenerateRequest>
```

Step 2 - Request (Output XML Document): Returned by Windcave PX Pay 2.0 API for merchant to use to redirect customer to Windcave Hosted Payments Page.

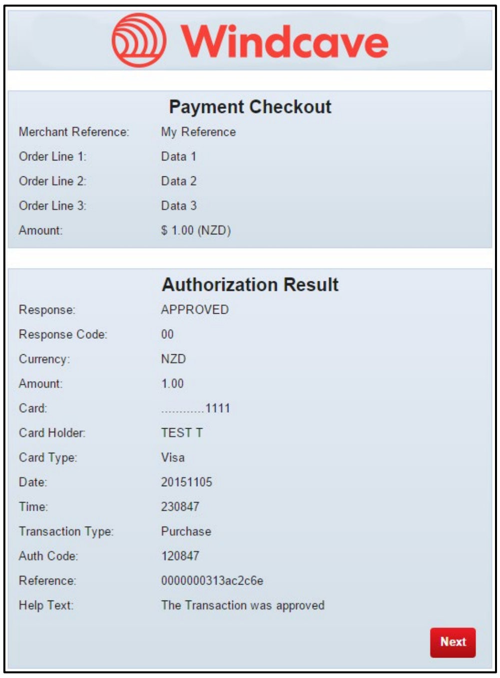
```
<Request valid="1">
<URI>https://sec.windcave.com/pxmi3/EF4054F622D6C4C1B4F9AEA59DC91CAD3654CD60ED7ED
04110CBC402959AC7CF035878AEB85D87223</URI>
</Request>
```

Step 3 – Redirection to HPP: Customer is redirected to the Hosted Payments Page to enter their credit card details

Page 1 – Hosted Payments Page (Enter CC details)



Page 2 - Transaction results



Note: Page 2 can be skipped so that redirection is done immediately after transaction is processed. Please refer to *HPP Customisation via Payline* for more information.

Step 4 – Redirection to UrlSuccess/UrlFail: Once transaction has been processed the customer is redirected back to the merchant's UrlSuccess of UrlFail depending on the transaction result. Transaction results are encrypted and returned in the "result" parameter of the URL string. You may also use the UrlCallback to receive the notification on your server side.

```
https://demo.windcave.com/SandboxSuccess.aspx?result=0000840000185416971799d38ef12345
&userid=SampleUserId
```

Step 5 - ProcessResponse (Input XML Document): Generated by the merchant using PxPayUserId, PxPaykey and Response details to decrypt the transaction result. The Response is the URI obtained from the "result" parameter in the URL string that is returned to the UrlSuccess/UrlFail page in step 4.